

How does a clinical trial work?

~~Cancer~~[®]

Clinical trials occur in four phases, and each phase has a different purpose.

Phase I



Focus on **safety** and the proper dose.

15 to 50 patients

Phase II



Focus on **effectiveness** and side effects.

Less than 100 patients

Phase III



Compares the **new treatment** to existing treatment.

Hundreds of people

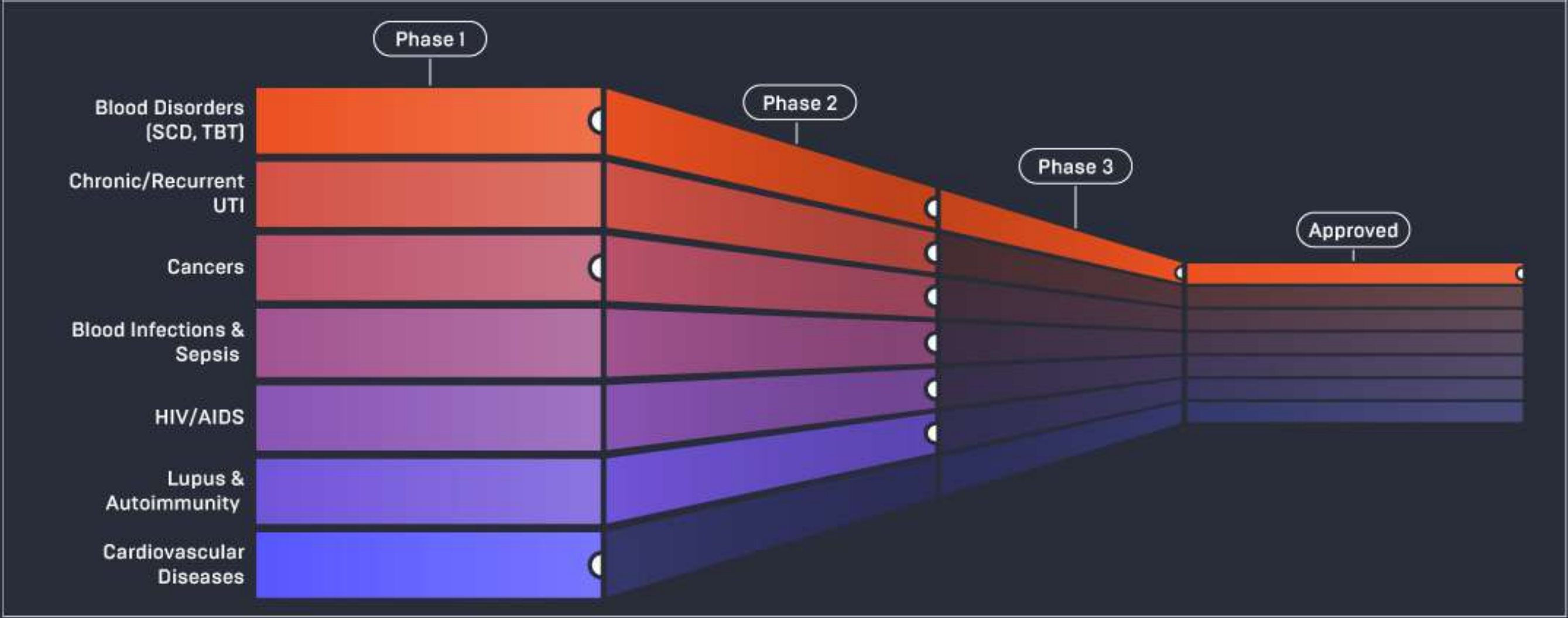
Phase IV

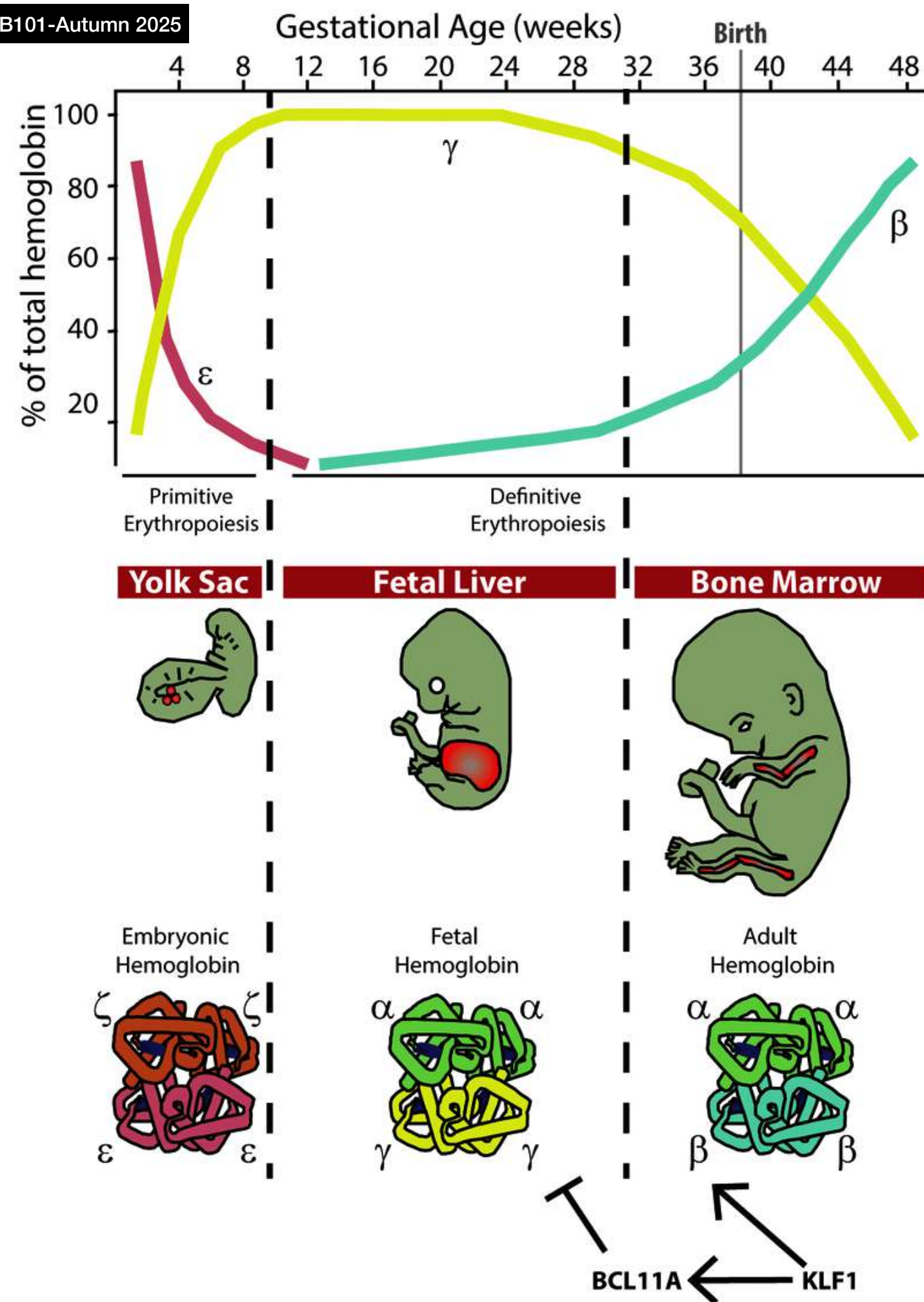


Treatment is **approved and available**. Long-term effects are observed.

Thousands of people

CRISPR Clinical Trials Progress - 2025





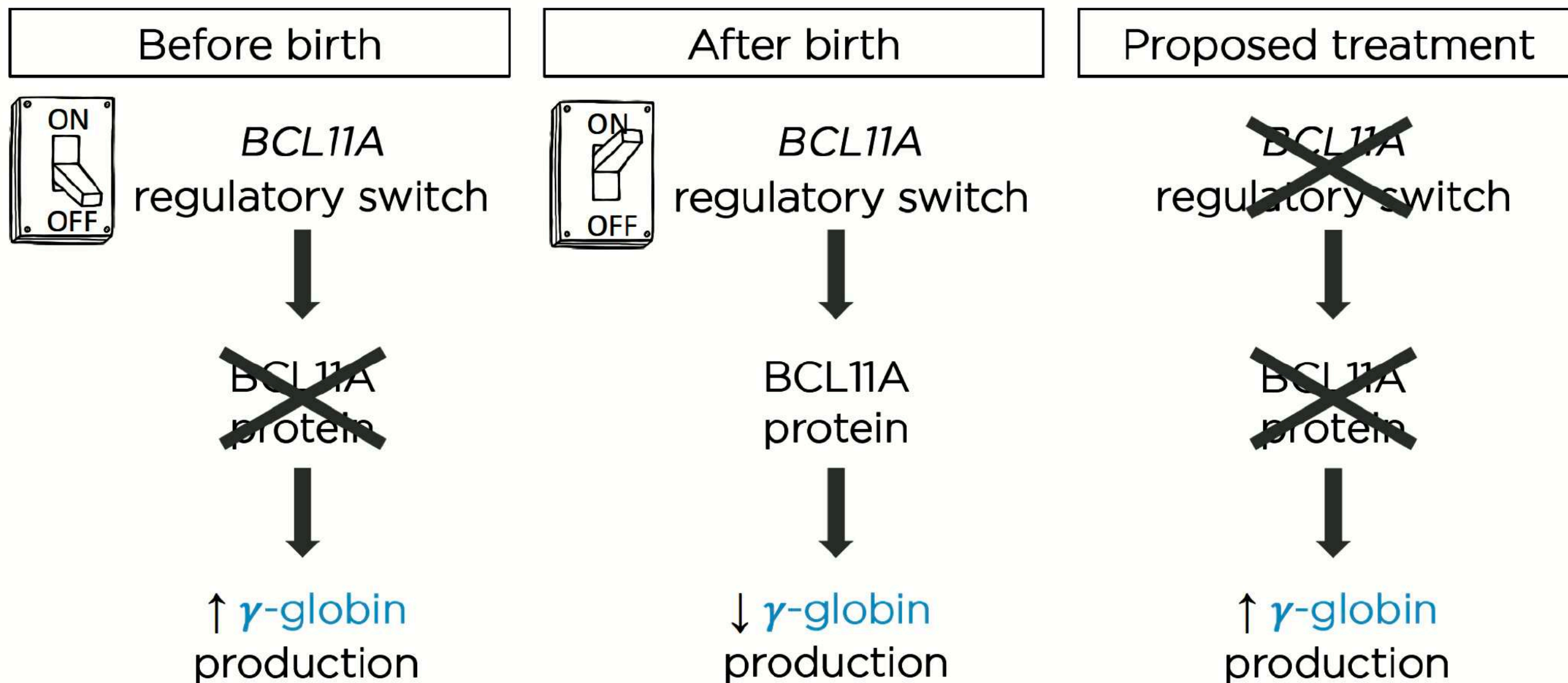
Hemoglobin Switch

- KLF1 activates β -globin and BCL11A
- BCL11A in turn represses γ -globin expression

To cure SCD, in which β -globin is mutated, what will you do?

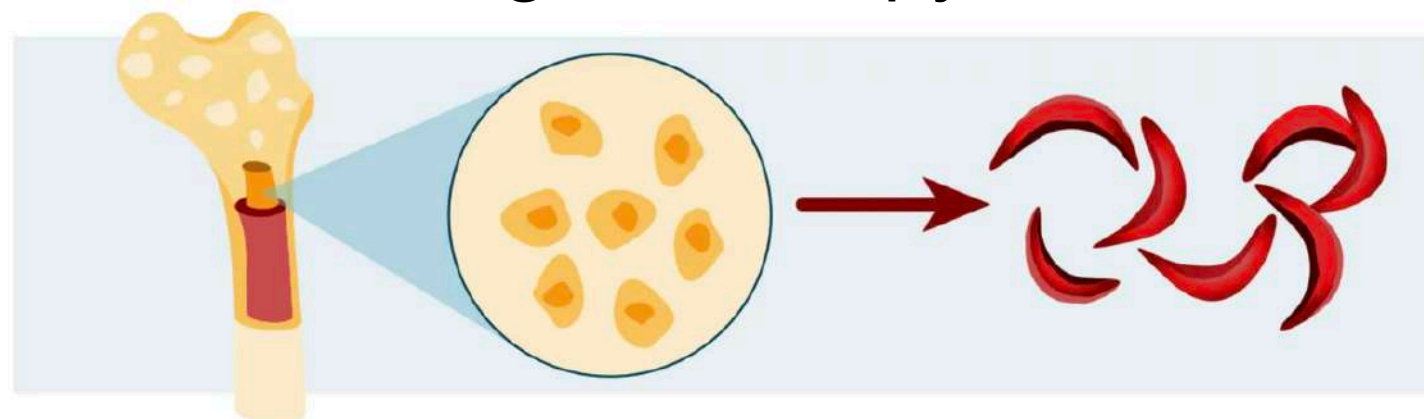
Casgevy

first approved CRISPR gene therapy for Sickle Cell Disease

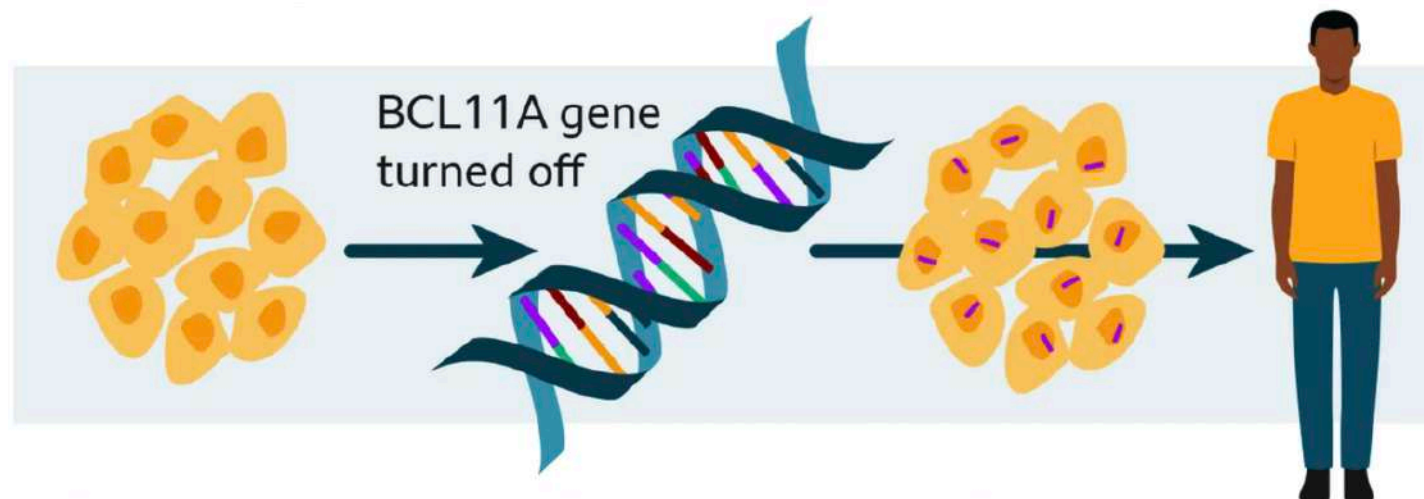


Casgevy

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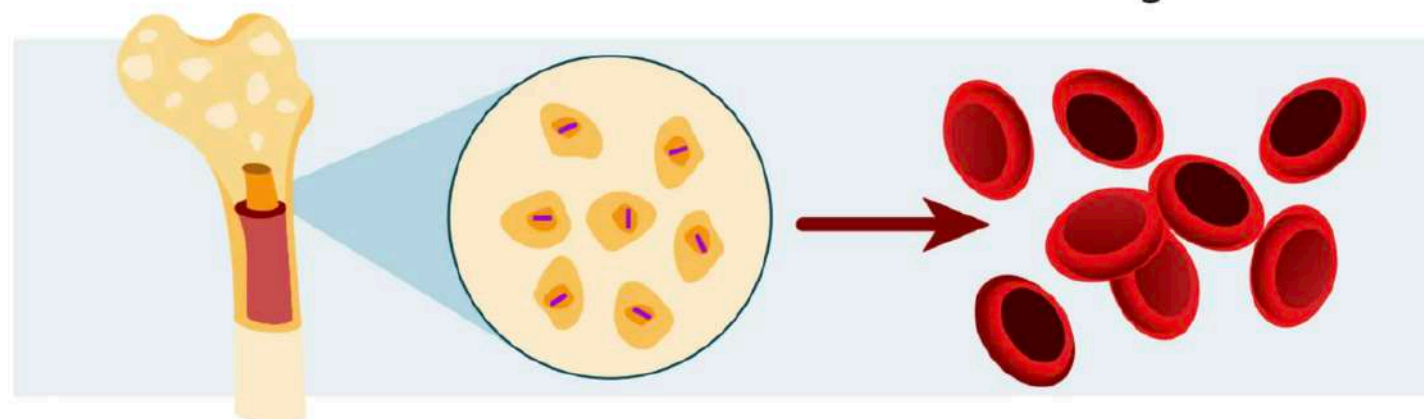
- 1 Jimi's stem cells in his bone marrow make diseased haemoglobin that can make red blood cells sickle-shaped



- 2 Stem cells extracted

- 3 Stem cells genetically modified

- 4 Genetically engineered stem cells given to Jimi



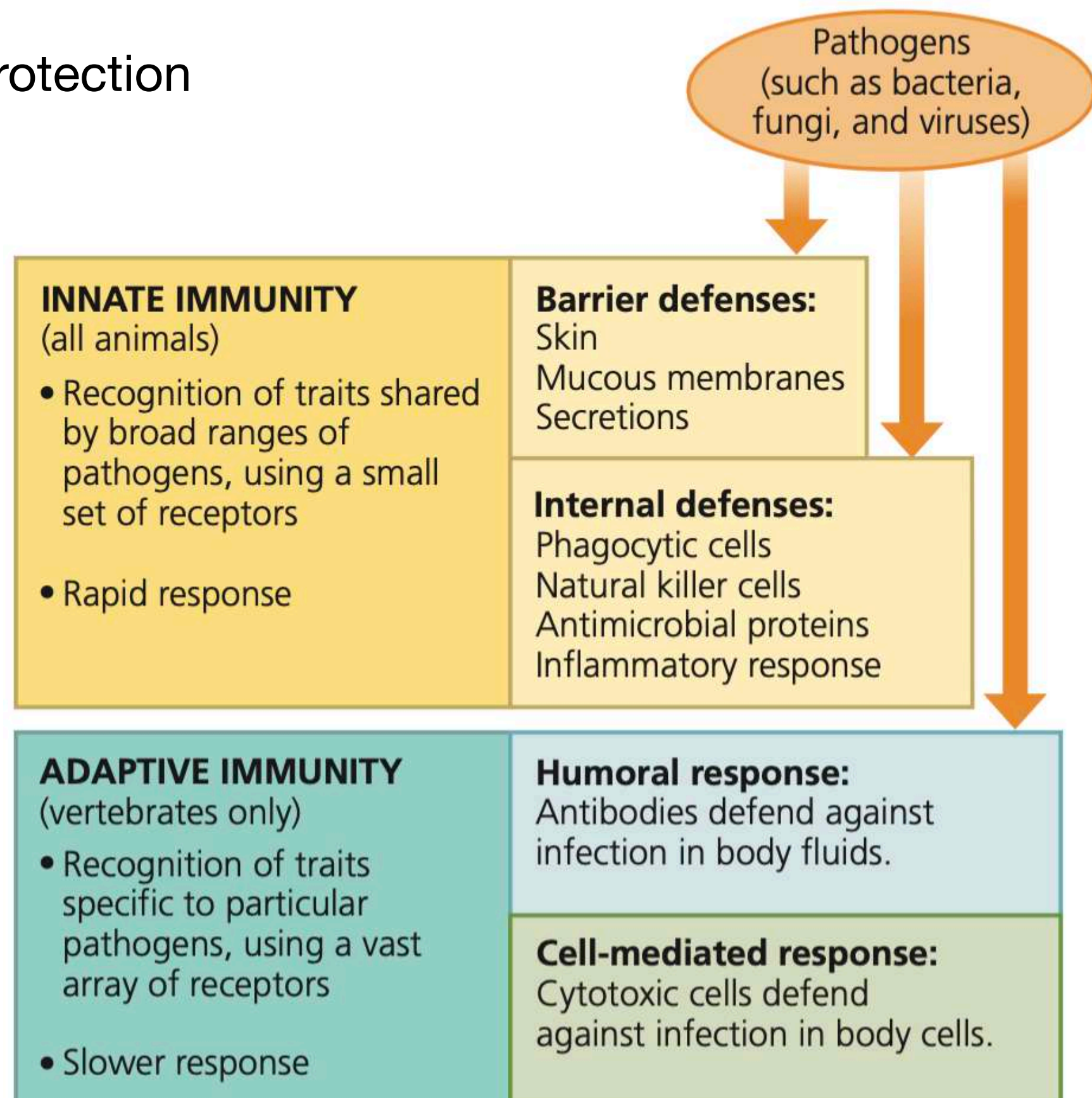
- 5 Engineered stem cells make healthy fetal haemoglobin and normal red blood cells

The Immune System

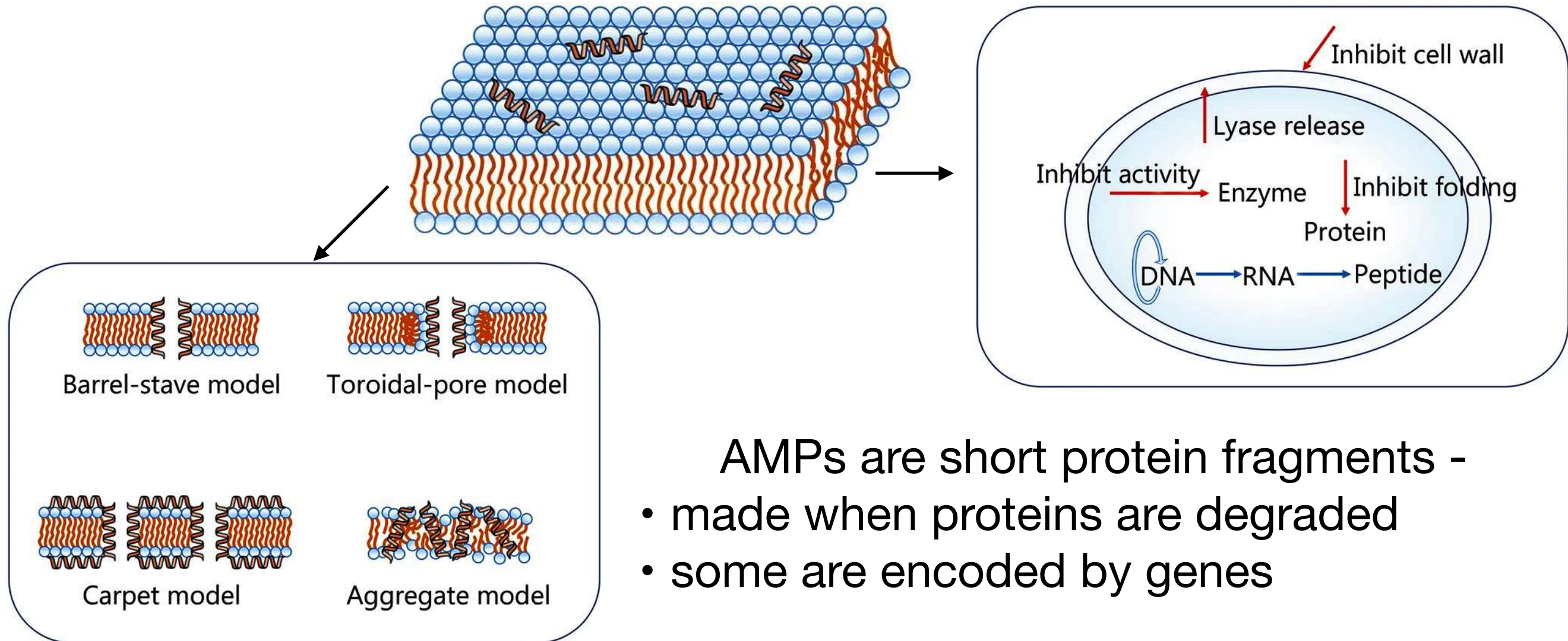
1. Multiple layers of protection
2. Prevent
3. Eliminate

Innate immunity = all animals and plants

Adaptive immunity = only in vertebrates



Innate Immunity - Antimicrobial peptides (AMPs)



AMPs are short protein fragments -

- made when proteins are degraded
- some are encoded by genes

- Upon infection with a pathogen, AMPs penetrate into the phospholipid membrane and disrupt it by interacting with the lipid bilayer
- This creates holes in the membrane of the pathogen and causes it to lyse
- AMPs can also interact with cellular proteins and nucleic acids inside the pathogen and disrupt their function
- AMPs are also produced by bacteria - to eliminate competition for a “niche”
- Bacterial AMPs in human gut are important in our immune system

Innate Immunity - Antimicrobial peptides (AMPs)

How specific are AMPs to one type of pathogen?

Experimental logic

Infect animals with a pathogen



Detect the AMP produced



Check that the AMP is able to neutralize that specific pathogen

Need the AMP to be genetically encoded



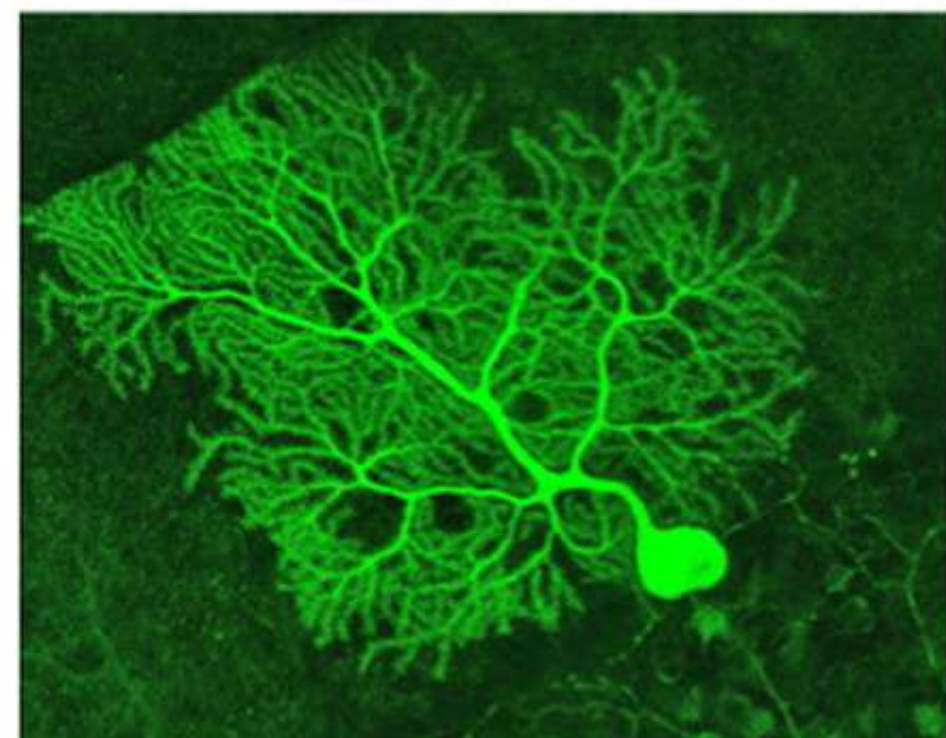
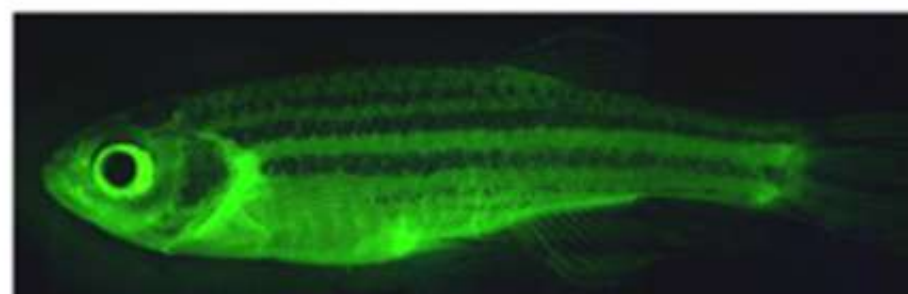
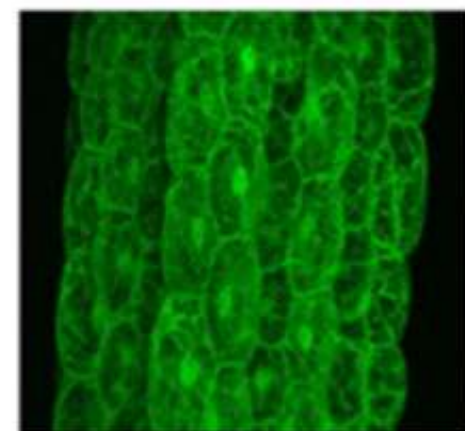
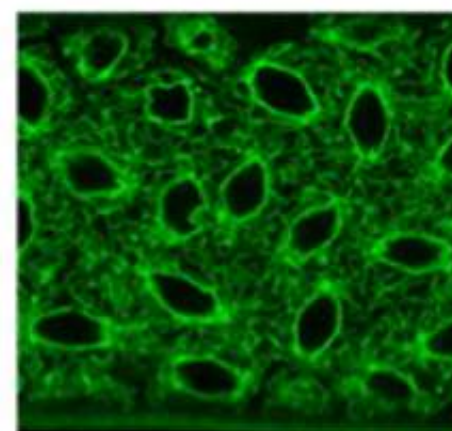
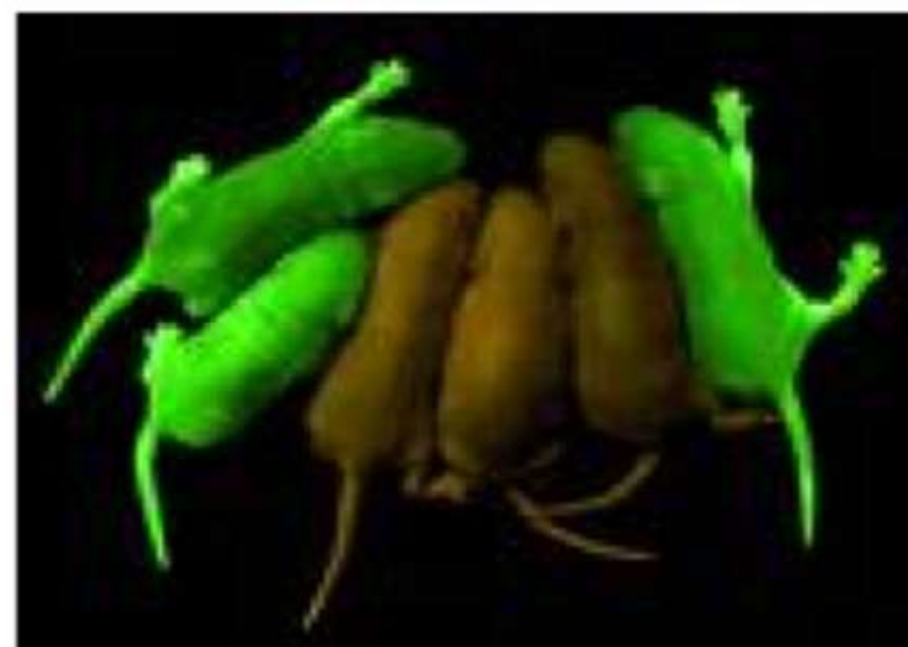
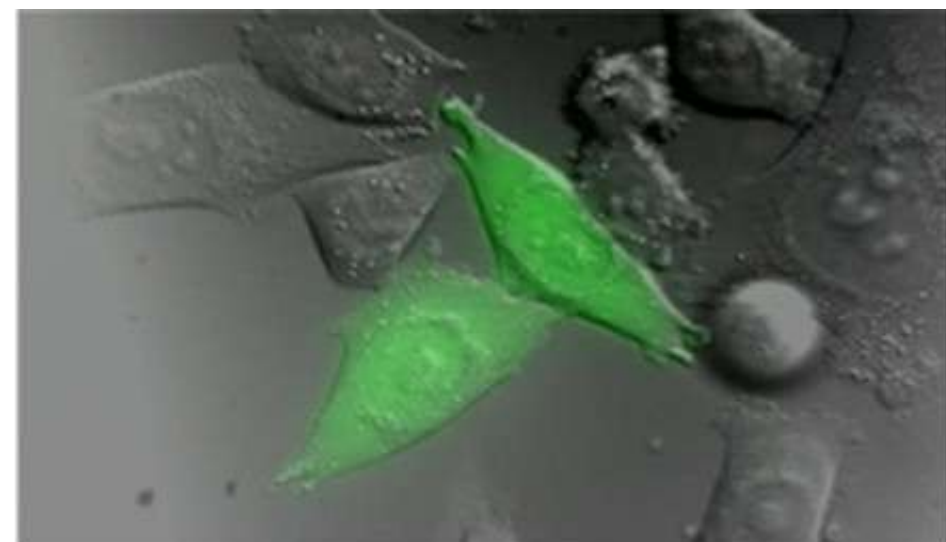
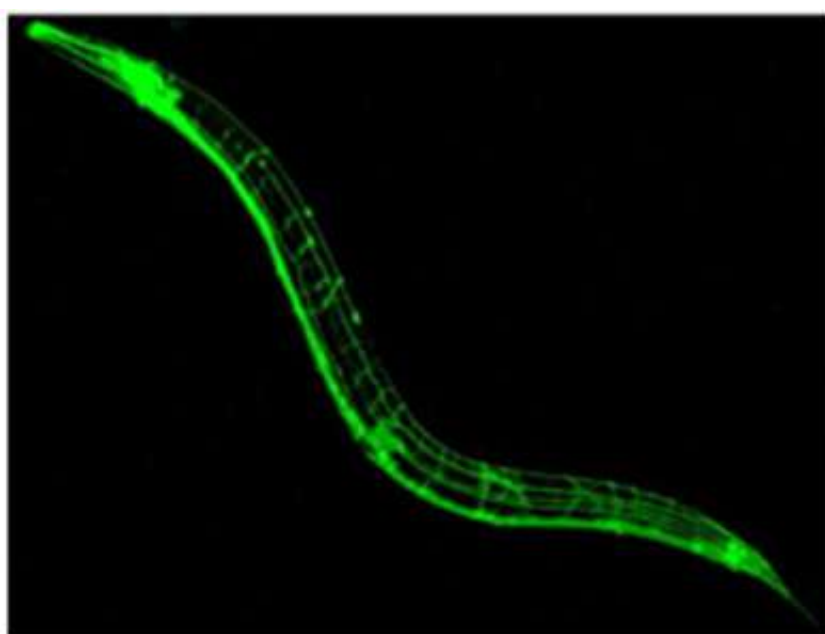
Allows one to “tag” the resultant AMP and “see” it

Green Fluorescent Protein and its variants

GFP is a genetically encoded fluorescent protein - has a high quantum yield
Variants can be made by mutating the original gene



Fluorescently tagging proteins in cells and animals



Innate Immunity - Antimicrobial peptides (AMPs)

How specific are AMPs to one type of pathogen?

Experimental logic

Infect animals with a pathogen



Fuse the AMP gene with GFP gene and “express” it in the animal



AMP is produced fused with GFP protein and will fluoresce when translated by cells in the animal



Infect the animal with pathogens



Check if the cells are fluorescing

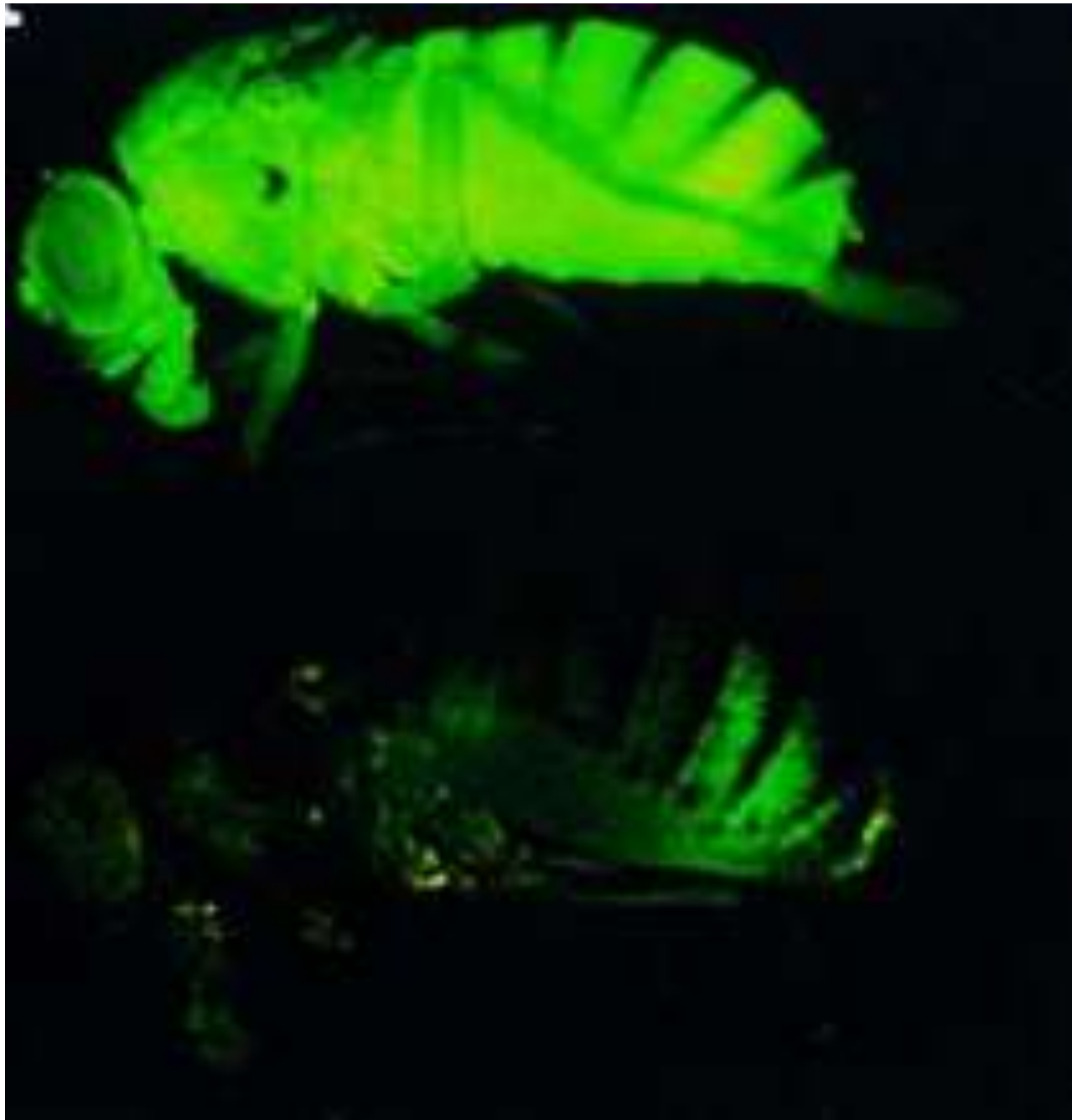


If fluorescence is detected = cells in the animal produced that AMP against the pathogen it was exposed to

Innate Immunity - Antimicrobial peptides (AMPs)

How specific are AMPs to one type of pathogen?

Done in *Drosophila* (fruit flies), using two bacteria as pathogens and two AMP genes as “read outs”



Pathogens used:
fungus *Neurospora crassa*
bacterium *Micrococcus luteus*

AMP genes GFP tagged:
droso mycin
defensin

Innate Immunity - Antimicrobial peptides (AMPs)

How specific are AMPs to one type of pathogen?

Experimental logic: must test two things - **specificity and function**

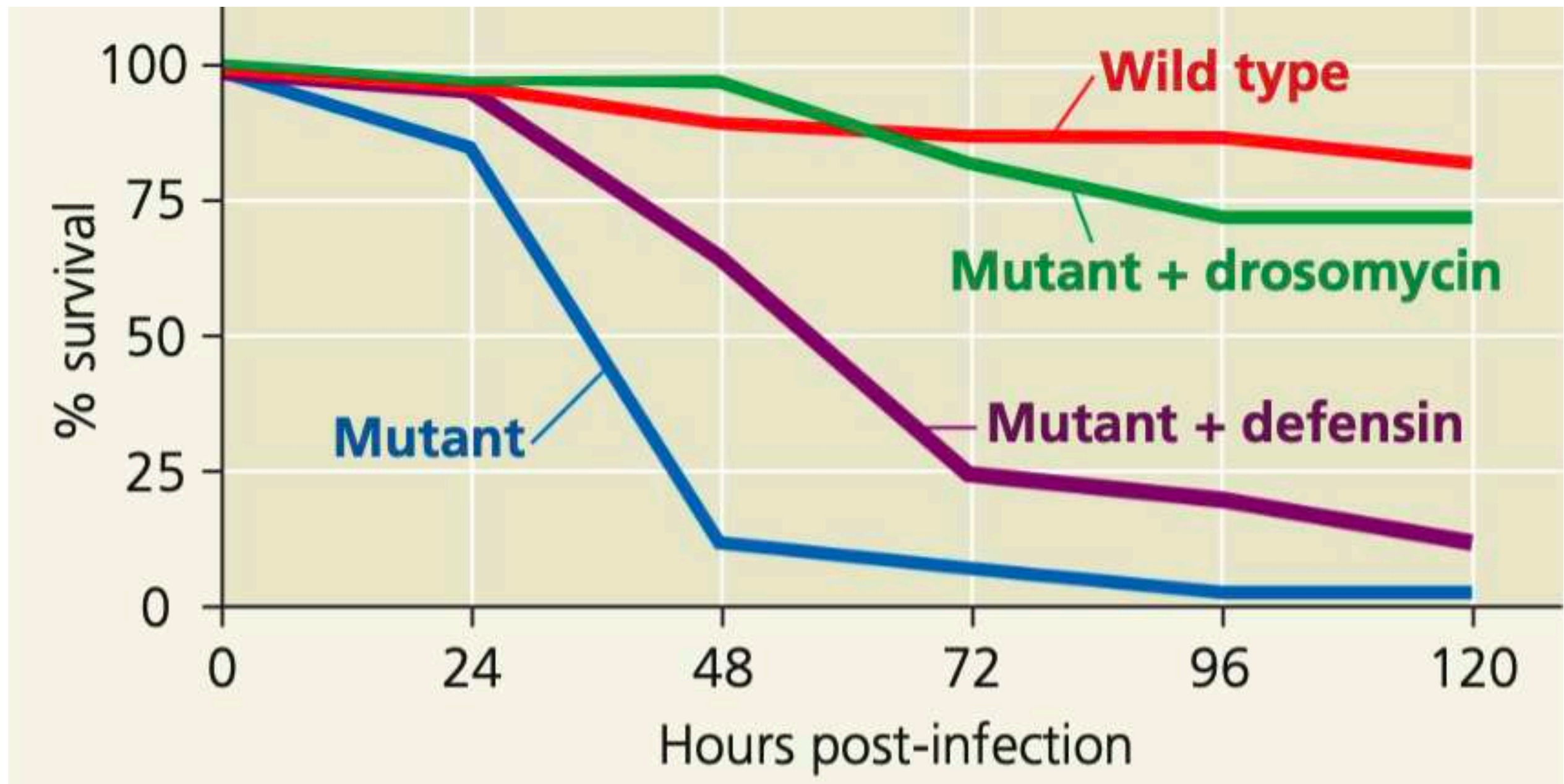
1. Generate mutant flies that cannot naturally produce AMPs - **allows you to create a functional assay**
2. The mutants will only make the AMPs you “express” in the flies, this will be the AMP::GFP fusion gene you have made - **allows you to control production of one AMP at a time**
3. Infect the mutant flies expressing one of the AMP::GFP fusion gene with one type of pathogen
4. AMP::GFP made in response to the pathogen can be seen by fluorescence - **allows you to check if the AMP made is unique for a pathogen**
5. Allow the infected flies to live. Survival of the flies will be the assay to test if the AMP produced can kill the pathogen - **this is the functional assay**

Logic loop complete

Innate Immunity - Antimicrobial peptides (AMPs)

How specific are AMPs to one type of pathogen?

Flies infected with fungus *Neurospora crassa*



What can you interpret and conclude from the graph?